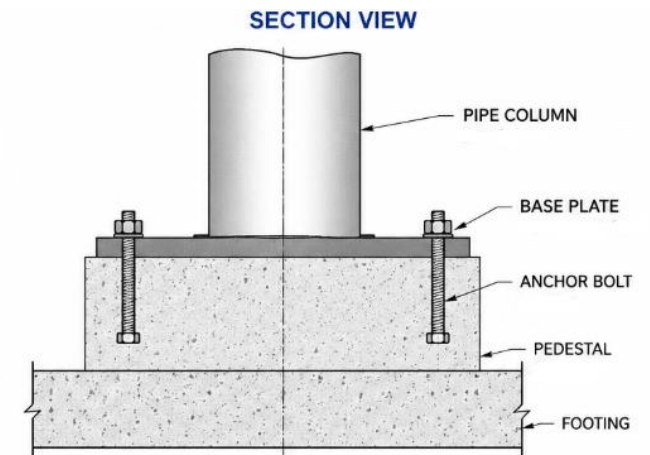
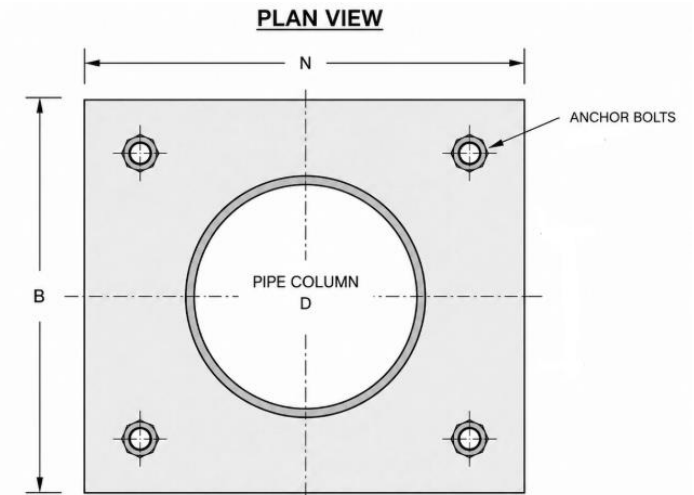


2. Base plate segment

Design Parameter	Symbol	Value	Source
Factored axial load (LRFD)	P_u	1609.756 kips	Main Tank Design (Predefined)
Governing overturning moment (LRFD)	M_u	5673.437 kip-ft	Main Tank Design (Predefined)
Pipe diameter	D_{pip}	20.04 in	Main Tank Design (Predefined)
Number of anchor bolts	N_b	4	From Anchor Bolt Design
Anchor bolt diameter	d_b	1.75 in	From Anchor Bolt Design
Bolt arrangement dimensions	$b \times l$	12 in $\times$ 12 in	From Anchor Bolt Design
Base-Plate cross-section	$B \times N$	30x30	
Pedestal cross-section	$P \times L$	39 in $\times$ 39 in	From Anchor Bolt Design
Concrete compressive strength	f'_c	4 ksi	User Input
Steel yield strength	F_y	36 ksi	User Input
Modulus of elasticity	E	29,000 ksi	AISC Material Property
LRFD resistance factor (plate bending)	ϕ_b	0.90	AISC LRFD
Required base plate thickness	t_{bp}	To be determined	AISC Base Plate Design



1. Determine the Factored Moment.

$$M_{u,ped} = \frac{M_u}{N_{ped}}$$

$$M_{u,ped} = \frac{5673.437}{4}$$

$$M_{u,ped} = 1418.36 \text{ kip-ft}$$

Convert to kip-in:

$$M_{u,ped} = 1418.36(12)$$

$$M_{u,ped} = 17020.31 \text{ kip-in}$$

2. Pedestal Area

$$A_2 = (39)(39)$$

$$A_2 = 1521 \text{ in}^2$$

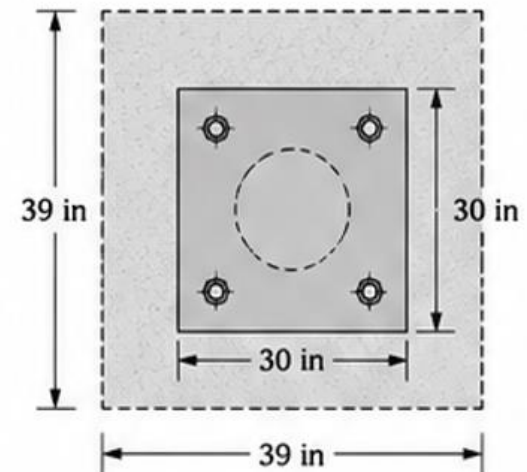
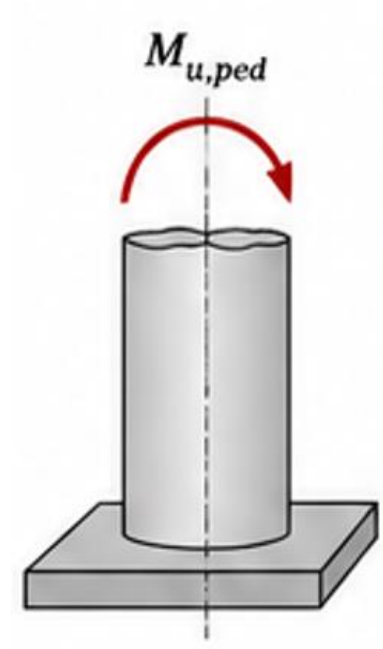
Base Plate Area

$$A_1 = (30)(30)$$

$$A_1 = 900 \text{ in}^2$$

Compute Bearing Stress

$$F_p = 0.85\phi_c f'_c \sqrt{\frac{A_2}{A_1}}$$



$$F_p = 0.85(0.65)(4) \sqrt{\frac{1521}{900}}$$

$$F_p = 2.87 \text{ ksi}$$

Check Limit

$$1.7\phi_c f'_c = 1.7(0.65)(4) = 4.42 \text{ ksi}$$

$$2.87 < 4.42 \dots \dots \dots \text{PASS}$$

3. Assume Base Plate Cross-Section

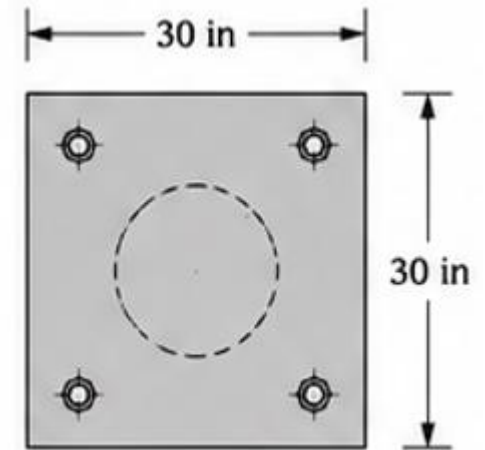
Adopt:

$$N = B = 30 \text{ in}$$

Base plate area:

$$A_1 = (30)(30)$$

$$A_1 = 900 \text{ in}^2$$



Check Bearing Capacity

$$P_n = F_p A_1$$

$$P_n = (2.87)(900)$$

$$P_n = 2583 \text{ kips}$$

Applied load:

$$P_u = 804.88 \text{ kips}$$

Note: for this scenario we considered half of the pedestals in compression and half of them in tension as show in the photo

$$2583 > 804.88$$

PASS

4. Determine Equivalent Eccentricity

$$e = \frac{M}{P}$$

Substitute:

$$e = \frac{17020.31}{402.44}$$

$$e = 42.29 \text{ in}$$

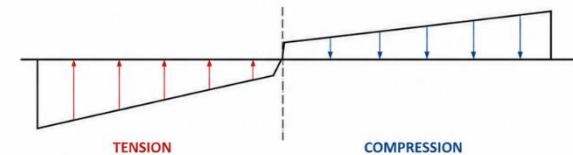
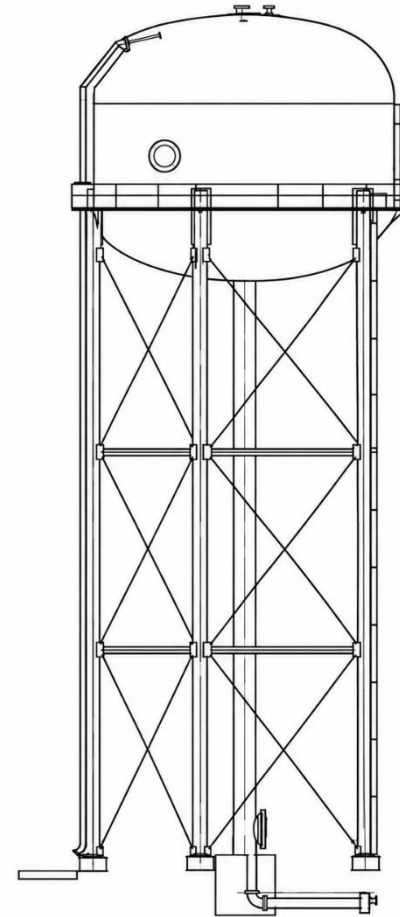
Check Bearing Condition

For full compression:

$$e \leq \frac{N}{6}$$

$$\frac{N}{6} = \frac{30}{6} = 5.0 \text{ in..... if this cause controls } P_u/(1.0*N)$$

Since:



$$42.29 > 5.0 \dots \text{if this cause controls } (P_u/0.5 \cdot N)$$

Full compression does not occur.

Therefore:

- Uplift develops.
- Anchor bolts resist tension.
- Partial compression exists beneath the base plate.

$$e = 42.29 \text{ in} > \frac{30}{6} = 5.0 \text{ in}$$

5. Determine Resultant Anchor Bolt Force

From the completed anchor bolt design:

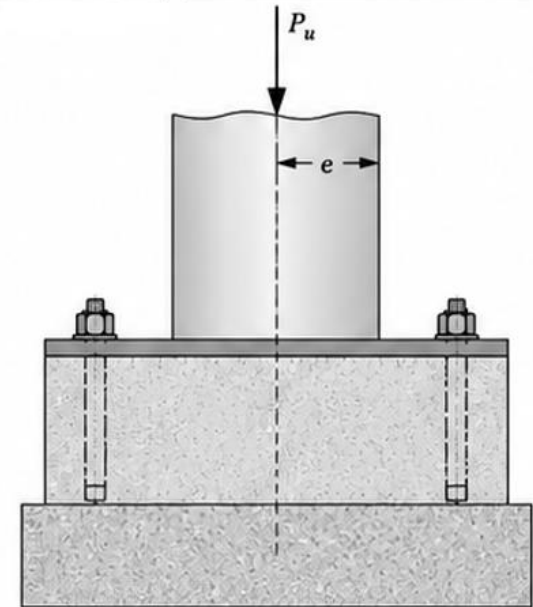
$$T_{\text{leg}} = 103.65 \text{ kips}$$

Number of anchor bolts:

$$N_b = 4$$

Anchor force per bolt:

$$T_{\text{bolt}} = \frac{T_{\text{leg}}}{N_b}$$



$$T_{\text{bolt}} = \frac{103.65}{4}$$

$$T_{\text{bolt}} = 25.91 \text{ kips}$$

Check Steel Tension Capacity

$$\phi T_s = 142.5 \text{ kips}$$

$$142.5 > 25.91$$

PASS

6. Determine Base Plate Thickness

6.1. Determine Bearing Area of Pipe Column

Cylinder diameter:

$$D = 1.67 \text{ ft}$$

$$D = 1.67(12)$$

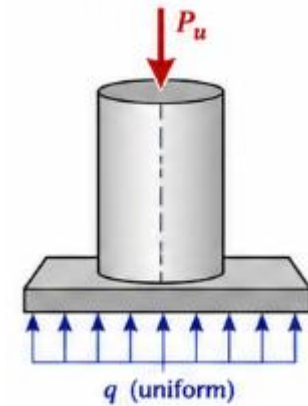
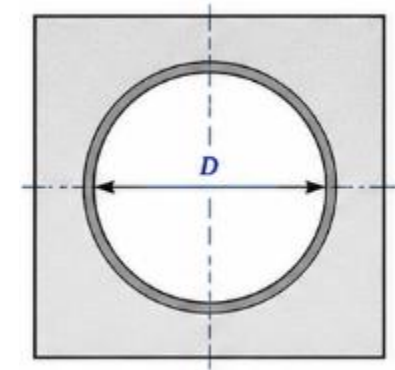
$$D = 20.04 \text{ in}$$

Column bearing area:

$$A_b = \frac{\pi D^2}{4}$$

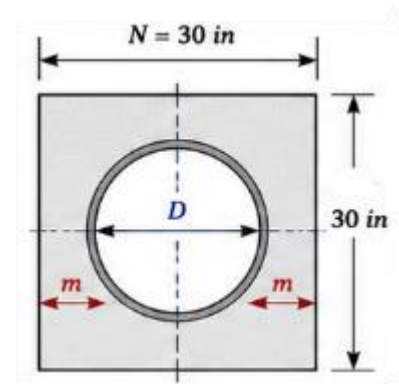
$$A_b = \frac{\pi(20.04)^2}{4}$$

$$A_b = 315.42 \text{ in}^2$$



6.2. Determine Bearing Pressure

$$q = \frac{P_u}{A_b}$$
$$q = \frac{402.44}{315.42}$$
$$q = 1.28 \text{ ksi}$$
$$\mathbf{q = 1.28 \text{ ksi}}$$

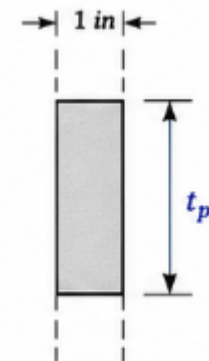
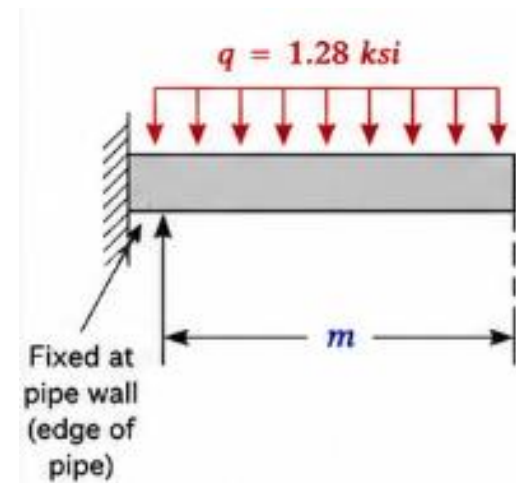


6.3. Determine Plate Projection

$$m = \frac{N - D}{2}$$
$$m = \frac{30 - 20.04}{2}$$
$$m = 4.98 \text{ in}$$
$$\mathbf{m = 4.98 \text{ in}}$$

6.4. Determine Plastic Moment per 1-in Strip

$$M_{plu} = \frac{qm^2}{2}$$
$$M_{plu} = \frac{(1.28)(4.98)^2}{2}$$
$$M_{plu} = 15.87 \text{ kip-in/in}$$
$$M_{plu} = 15.87 \text{ kip-in/in}$$



6.5. Determine Required Thickness

Using AISC LRFD:

$$t_p = \sqrt{\frac{4M_{plu}}{0.90F_y}}$$

$$t_p = \sqrt{\frac{4(15.87)}{0.90(36)}}$$

$$t_p = \sqrt{\frac{63.48}{32.40}}$$

$$t_p = \sqrt{1.959}$$

$$t_p = 1.40 \text{ in}$$

$$\mathbf{t_p = 1.40 \text{ in}}$$

Adopt:

$$\mathbf{\underline{t_p = 1.50 \text{ in}}}$$

